

An empirical example: Four terminals and two areas

Daniel R. Miranda-Esquivel

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```
# Four taxa and two areas
```

```
## Libraries
```

```
library(blepd)
```

```
## Loading required package: ape
```

```
## Loading required package: picante
```

```
## Loading required package: vegan
```

```
## Loading required package: permute
```

```
##
```

```
## Attaching package: 'permute'
```

```
## The following object is masked from 'package:devtools':
```

```
##
```

```
##      check
```

```
## Loading required package: lattice
```

```
## This is vegan 2.6-6.1
```

```
## Loading required package: nlme
```

```
library(ggtree)
```

```
## ggtree v3.10.1 For help: https://yulab-smu.top/treedata-book/
```

```
##
```

```
## If you use the ggtree package suite in published research, please cite
```

```
## the appropriate paper(s):
```

```
##
```

```
## Guangchuang Yu, David Smith, Huachen Zhu, Yi Guan, Tommy Tsan-Yuk Lam.
```

```
## ggtree: an R package for visualization and annotation of phylogenetic
```

```
## trees with their covariates and other associated data. Methods in
```

```
## Ecology and Evolution. 2017, 8(1):28-36. doi:10.1111/2041-210X.12628
```

```
##
```

```
## Shuangbin Xu, Lin Li, Xiao Luo, Meijun Chen, Wenli Tang, Li Zhan, Zehan
```

```
## Dai, Tommy T. Lam, Yi Guan, Guangchuang Yu. Ggtree: A serialized data
```

```
## object for visualization of a phylogenetic tree and annotation data.
```

```
## iMeta 2022, 1(4):e56. doi:10.1002/imt2.56
```

```
##
```

```
## LG Wang, TTY Lam, S Xu, Z Dai, L Zhou, T Feng, P Guo, CW Dunn, BR
```

```
## Jones, T Bradley, H Zhu, Y Guan, Y Jiang, G Yu. treeio: an R package
```

```

## for phylogenetic tree input and output with richly annotated and
## associated data. Molecular Biology and Evolution. 2020, 37(2):599-603.
## doi: 10.1093/molbev/msz240

##
## Attaching package: 'ggtree'

## The following object is masked from 'package:nlme':
##
##      collapse

## The following object is masked from 'package:ape':
##
##      rotate

library(ggplot2)

## Data

# Call the 4 terminals example

##data(package = "blepd")

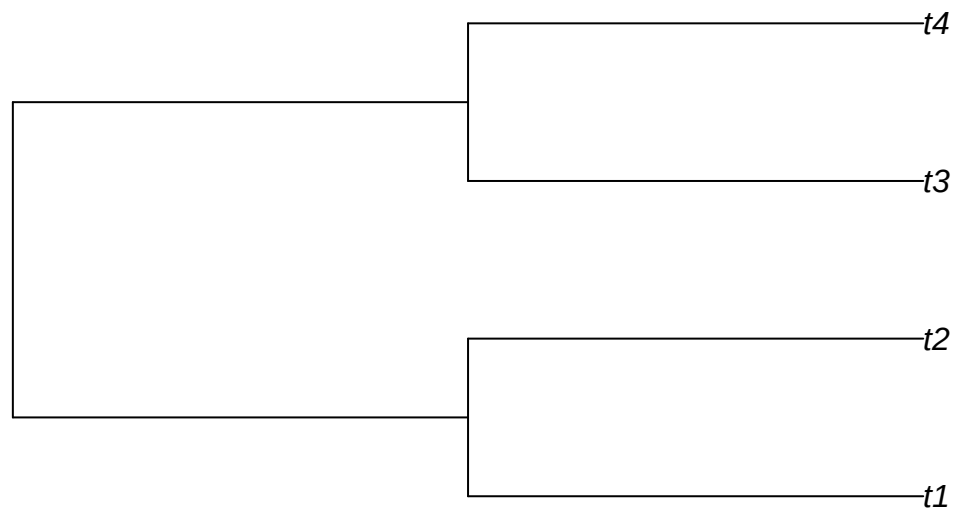
## trees

data(tree)

#str(tree)

plot(tree,use.edge.length =T)

```



```

initialTree <- tree

## distributions

data(distribution)

##str(distribution)

dist4taxa <- distribution

## distribution to XY to be able to plot the data, otherwise, skip the next step

distXY <- matrix2XY(dist4taxa)

distXY

## Terminal Area
## 1      t1    A1
## 2      t3    A1
## 3      t2    A2
## 4      t4    A2

## plotting

## the tree

```

```

plotTree <- ggtree(initialTree, ladderize=TRUE,
                   color="black", size=0.5, linetype="dotted") +
  geom_tiplab(size=4.5, color="black") +
  theme_tree2() +
  ggtitle("Four terminals, all branches are == 1")

##
print(plotTree)

```

Four terminals, all branches are == 1



```

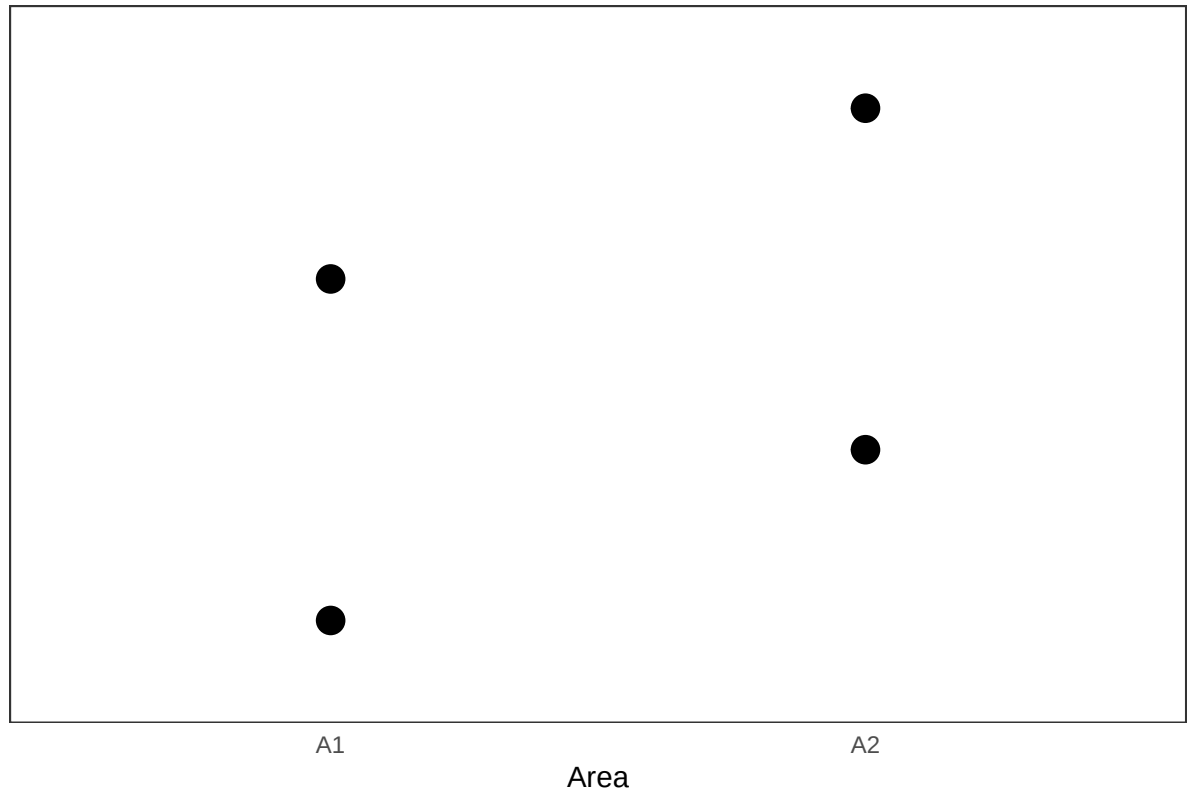
## the distribution

plotDistrib <- ggplot(data=distXY,
                      aes(x= Area, y= Terminal , size =150)) +
  geom_point() +
  theme_bw() +
  theme( axis.text.y = element_blank(),
        panel.grid.major = element_blank(), # Remove major gridlines
        panel.grid.minor = element_blank(), # Remove minor gridlines
        axis.line = element_line(colour = NA_character_), # Remove axis lines
        axis.ticks = element_blank(), # Remove axis ticks
        legend.position = "none"
  ) +
  labs(title = "Distributions",
       y = "",

```

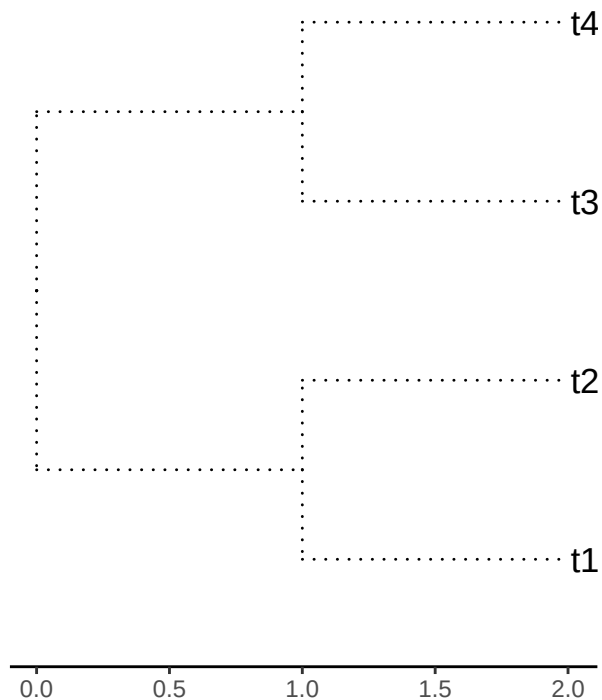
```
x = "Area")  
  
##  
print(plotDistrib)
```

Distributions

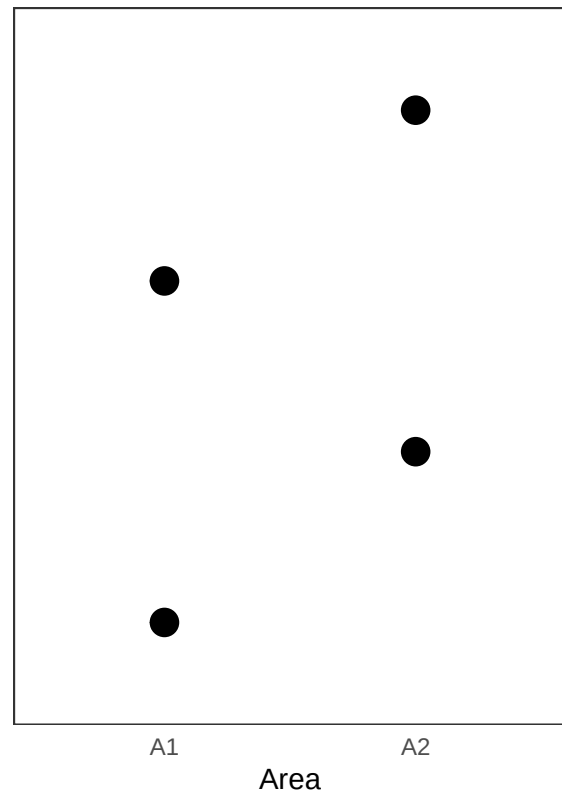


```
cowplot::plot_grid(plotTree, plotDistrib, ncol=2)
```

Four terminals, all branches are == 1



Distributions



```
#We check whether names in both objects: initialTree and dist4taxa, are the same.
```

```
##all(colnames(dist4taxa) == initialTree$tip.label)
```

```
all(colnames(dist4taxa) %in% initialTree$tip.label)
```

```
## [1] TRUE
```

```
#We report the branch length, and calculate the PD values.
```

```
initialTree$edge.length
```

```
## [1] 1 1 1 1 1 1
```

```
initialPD <- PDindex(tree=initialTree, distribution = dist4taxa)
```

```
initialPD
```

```
## [1] 4 4
```

```
#As expected there is a tie between both areas.
```

```
#Now, we can see the impact of swapping the branch lengths; we will use the three  
#available models to swap: "simpleswap", "allswap", "uniform", and we will swap the  
#three classes of branches: "terminals", "internals", "all".
```

```
##?swapBL
```

```
for( modelo in c("simplswap","allswap","uniform") ){  
  for( rama in c("terminals","internals","all") ){  
    ## cat( "\n\t Branchs swapped=\t",rama,".\t",sep="" )  
    val <- swapBL(tree=initialTree,  
                  distribution = dist4taxa,  
                  model = modelo,  
                  branch = rama  
                  )  
    printSwapBL(val, compact=TRUE)  
  
    cat( "\n\n" )  
  }  
}
```

```
## model to test simplswap reps 100  
## A1A2  
## 1 100  
##  
##  
## model to test simplswap reps 100  
## A1A2  
## 1 100  
##  
##  
## model to test simplswap reps 100  
## A1A2  
## 1 100  
##  
##  
## model to test allswap reps 100  
## A1A2  
## 1 100  
##  
##  
## model to test allswap reps 100  
## A1A2  
## 1 100  
##  
##  
## model to test allswap reps 100  
## A1A2  
## 1 100  
##  
##  
## model to test uniform reps 100
```

```
## A1A2
## 1 100
##
##
## model to test uniform reps 100
## A1A2
## 1 100
##
##
## model to test uniform reps 100
## A1A2
## 1 100
```

#As expected, as all braches are EQUAL nothing happens

#Now, let us see if the results could be related to the branch length (or not).

*## we could use a single command (*evalBranch*)*

##?evalBranch

```
AllTerminals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="terminals",
                           approach="all")
```

Loading required package: maps

##

Attaching package: 'phytools'

The following object is masked from 'package:vegan':

##

scores

let's see the structure of the output

##str(AllTerminals)

Now we can print a more friendly output

```
printEvalBranch(AllTerminals, compact=FALSE)
```

```
## node initialArea FinalArea Aproach %Delta
## 1 1 A1A2 A2 lower -1.024
## 2 2 A1A2 A1 lower -1.024
## 3 3 A1A2 A2 lower -1.024
## 4 4 A1A2 A1 lower -1.024
## 5 1 A1A2 A1 upper 1.062
## 6 2 A1A2 A2 upper 1.062
## 7 3 A1A2 A1 upper 1.062
```

```

## 8      4      A1A2      A2      upper      1.062
## Now the internals

AllInternals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="internals",
                           approach="all")

printEvalBranch(AllInternals, compact=FALSE)

##      node initialArea FinalArea Aproach %Delta
## 1      6      A1A2      A1A2      lower      0
## 2      7      A1A2      A1A2      lower      0
## 3      6      A1A2      A1A2      upper      0
## 4      7      A1A2      A1A2      upper      0
##now with a longer branch

initialTree <- tree

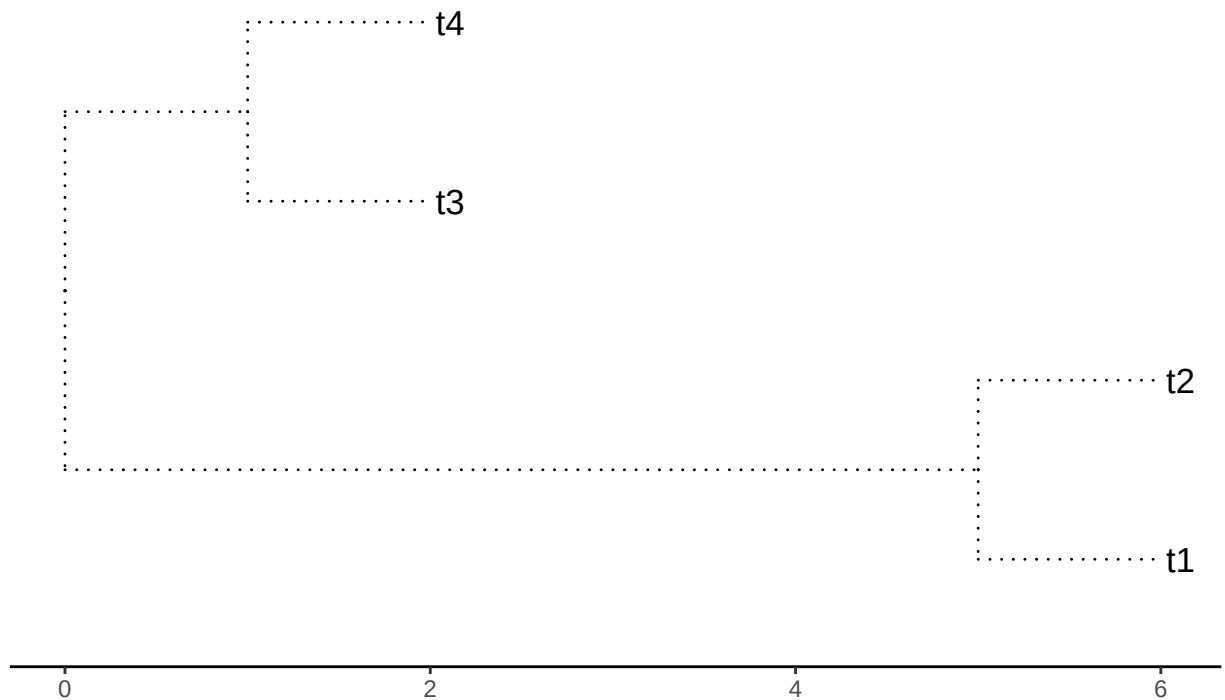
initialTree$edge.length[1] <- 5

plotTree <- ggtree(initialTree, ladderize=TRUE,
                   color="black", size=0.5, linetype="dotted") +
  geom_tiplab(size=4.5, color="black") +
  theme_tree2() +
  ggtitle(paste("Four terminals. Branches 1=",initialTree$edge.length[1],
               ". 2=",initialTree$edge.length[2],".",sep=""))

##
print(plotTree)

```

Four terminals. Branches 1=5. 2=1.



```
for( modelo in c("simpleswap","allswap","uniform") ){
  for( rama in c("terminals","internals","all") ){
    ## cat( "\n\t Branchs swapped=\t",rama,".\t",sep="" )
    val <- swapBL(tree=initialTree,
                  distribution = dist4taxa,
                  model = modelo,
                  branch = rama
                )
    printSwapBL(val, compact=TRUE)

    cat( "\n\n" )
  }
}
```

```
## model to test simpleswap reps 100
## A1A2
## 1 100
##
##
## model to test simpleswap reps 100
## A1A2
## 1 100
```

```

##
##
## model to test simpleswap reps 100
##   A1 A1A2 A2
## 1 17   70 13
##
##
## model to test allswap reps 100
##   A1A2
## 1   100
##
##
## model to test allswap reps 100
##   A1A2
## 1   100
##
##
## model to test allswap reps 100
##   A1 A1A2 A2
## 1 40   36 24
##
##
## model to test uniform reps 100
##   A1A2
## 1   100
##
##
## model to test uniform reps 100
##   A1A2
## 1   100
##
##
## model to test uniform reps 100
##   A1 A2
## 1 47 53

AllTerminals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="terminals",
                           approach="all")

printEvalBranch(AllTerminals, compact=FALSE)

##   node initialArea FinalArea Aproach %Delta
## 1    1          A1A2        A2   lower -1.024
## 2    2          A1A2        A1   lower -1.024
## 3    3          A1A2        A2   lower -1.024
## 4    4          A1A2        A1   lower -1.024
## 5    1          A1A2        A1   upper  1.055
## 6    2          A1A2        A2   upper  1.055
## 7    3          A1A2        A1   upper  1.055
## 8    4          A1A2        A2   upper  1.055

## Now the internals

```

```

AllInternals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="internals",
                           approach="all")

printEvalBranch(AllInternals, compact=FALSE)

##   node initialArea FinalArea Aproach %Delta
## 1    6          A1A2      A1A2  lower    0
## 2    7          A1A2      A1A2  lower    0
## 3    6          A1A2      A1A2  upper    0
## 4    7          A1A2      A1A2  upper    0
#now with a longer branch

initialTree <- tree

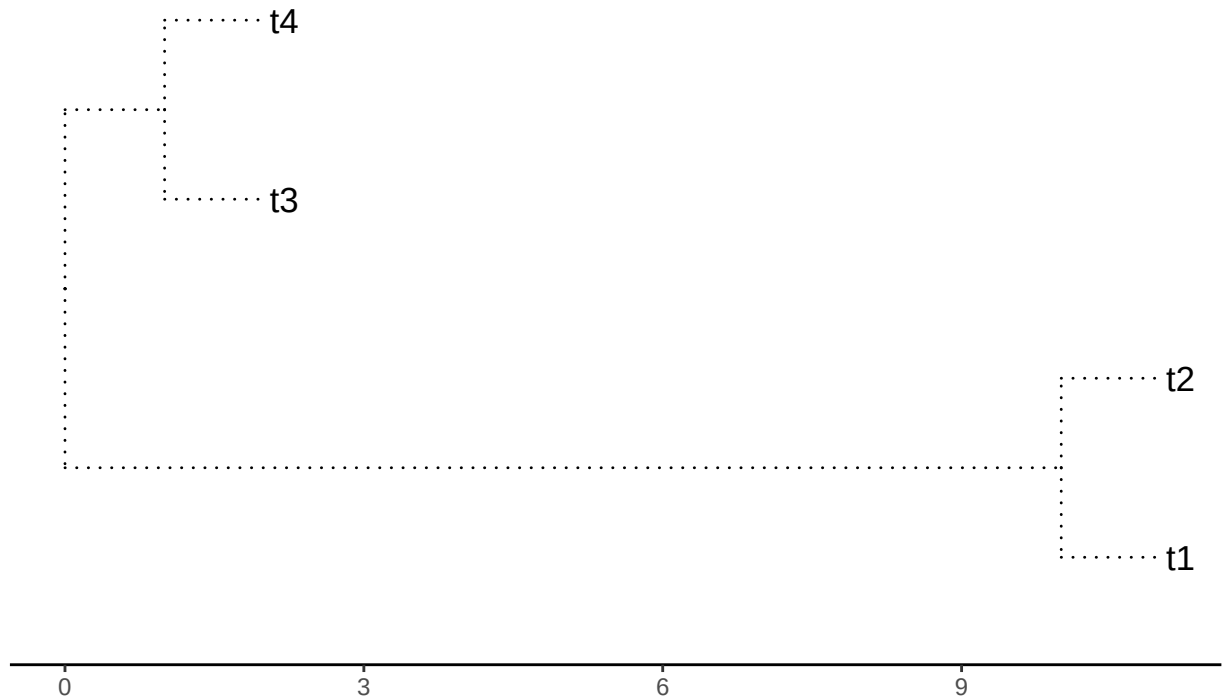
initialTree$edge.length[1] <- 10

plotTree <- ggtree(initialTree, ladderize=TRUE,
                   color="black", size=0.5, linetype="dotted") +
  geom_tiplab(size=4.5, color="black") +
  theme_tree2() +
  ggtitle(paste("Four terminals. Branches 1=",initialTree$edge.length[1],
               ". 2=",initialTree$edge.length[2],".",sep=""))

##
print(plotTree)

```

Four terminals. Branches 1=10. 2=1.



```
for( modelo in c("simpleswap","allswap","uniform") ){
  for( rama in c("terminals","internals","all") ){
    ## cat( "\n\t Branchs swapped=\t",rama,".\t",sep="" )
    val <- swapBL(tree=initialTree,
                  distribution = dist4taxa,
                  model = modelo,
                  branch = rama
                )
    printSwapBL(val, compact=TRUE)

    cat( "\n\n" )
  }
}
```

```
## model to test simpleswap reps 100
## A1A2
## 1 100
##
##
## model to test simpleswap reps 100
## A1A2
## 1 100
```

```

##
##
## model to test simpleswap reps 100
##   A1 A1A2 A2
## 1 10   76 14
##
##
## model to test allswap reps 100
##   A1A2
## 1   100
##
##
## model to test allswap reps 100
##   A1A2
## 1   100
##
##
## model to test allswap reps 100
##   A1 A1A2 A2
## 1 25   37 38
##
##
## model to test uniform reps 100
##   A1A2
## 1   100
##
##
## model to test uniform reps 100
##   A1A2
## 1   100
##
##
## model to test uniform reps 100
##   A1 A2
## 1 49 51

AllTerminals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="terminals",
                           approach="all")

printEvalBranch(AllTerminals, compact=FALSE)

##   node initialArea FinalArea Aproach %Delta
## 1    1          A1A2        A2   lower -1.024
## 2    2          A1A2        A1   lower -1.024
## 3    3          A1A2        A2   lower -1.024
## 4    4          A1A2        A1   lower -1.024
## 5    1          A1A2        A1   upper  1.022
## 6    2          A1A2        A2   upper  1.022
## 7    3          A1A2        A1   upper  1.022
## 8    4          A1A2        A2   upper  1.022

## Now the internals

```

```

AllInternals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="internals",
                           approach="all")

printEvalBranch(AllInternals, compact=FALSE)

##   node initialArea FinalArea Aproach %Delta
## 1    6          A1A2      A1A2   lower     0
## 2    7          A1A2      A1A2   lower     0
## 3    6          A1A2      A1A2   upper     0
## 4    7          A1A2      A1A2   upper     0
#now with a longer branch

initialTree <- tree

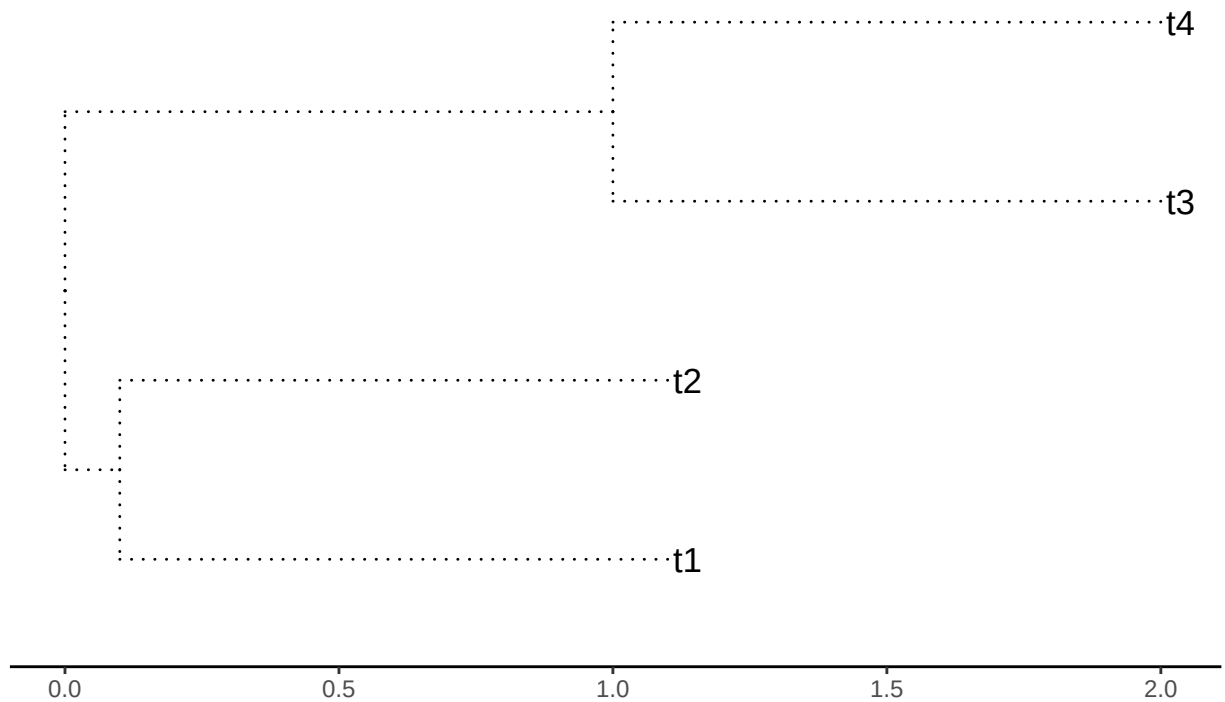
initialTree$edge.length[1] <- 0.1

plotTree <- ggtree(initialTree, ladderize=TRUE,
                   color="black", size=0.5, linetype="dotted") +
  geom_tiplab(size=4.5, color="black") +
  theme_tree2() +
  ggtitle(paste("Four terminals. Branches 1=",initialTree$edge.length[1],
               ". 2=",initialTree$edge.length[2],".",sep=""))

##
print(plotTree)

```

Four terminals. Branches 1=0.1. 2=1.



```
for( modelo in c("simpleswap","allswap","uniform") ){
  for( rama in c("terminals","internals","all") ){
    ## cat( "\n\t Branchs swapped=\t",rama,".\t",sep="" )
    val <- swapBL(tree=initialTree,
                  distribution = dist4taxa,
                  model = modelo,
                  branch = rama
                )
    printSwapBL(val, compact=TRUE)

    cat( "\n\n" )
  }
}
```

```
## model to test simpleswap reps 100
## A1A2
## 1 100
##
##
## model to test simpleswap reps 100
## A1A2
## 1 100
```

```

##
##
## model to test simpleswap reps 100
##   A1 A1A2 A2
## 1 14   73 13
##
##
## model to test allswap reps 100
##   A1A2
## 1   100
##
##
## model to test allswap reps 100
##   A1A2
## 1   100
##
##
## model to test allswap reps 100
##   A1 A1A2 A2
## 1 37   37 26
##
##
## model to test uniform reps 100
##   A1A2
## 1   100
##
##
## model to test uniform reps 100
##   A1A2
## 1   100
##
##
## model to test uniform reps 100
##   A1 A2
## 1 53 47

AllTerminals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="terminals",
                           approach="all")

printEvalBranch(AllTerminals, compact=FALSE)

##   node initialArea FinalArea Aproach %Delta
## 1    1          A1A2        A2   lower -1.024
## 2    2          A1A2        A1   lower -1.024
## 3    3          A1A2        A2   lower -1.024
## 4    4          A1A2        A1   lower -1.024
## 5    1          A1A2        A1   upper  1.051
## 6    2          A1A2        A2   upper  1.051
## 7    3          A1A2        A1   upper  1.051
## 8    4          A1A2        A2   upper  1.051

## Now the internals

```

```

AllInternals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="internals",
                           approach="all")

printEvalBranch(AllInternals, compact=FALSE)

##   node initialArea FinalArea Approach %Delta
## 1    6          A1A2      A1A2   lower     0
## 2    7          A1A2      A1A2   lower     0
## 3    6          A1A2      A1A2   upper     0
## 4    7          A1A2      A1A2   upper     0
#now with a longer branch in brach 2

#now with a longer branch

initialTree <- tree

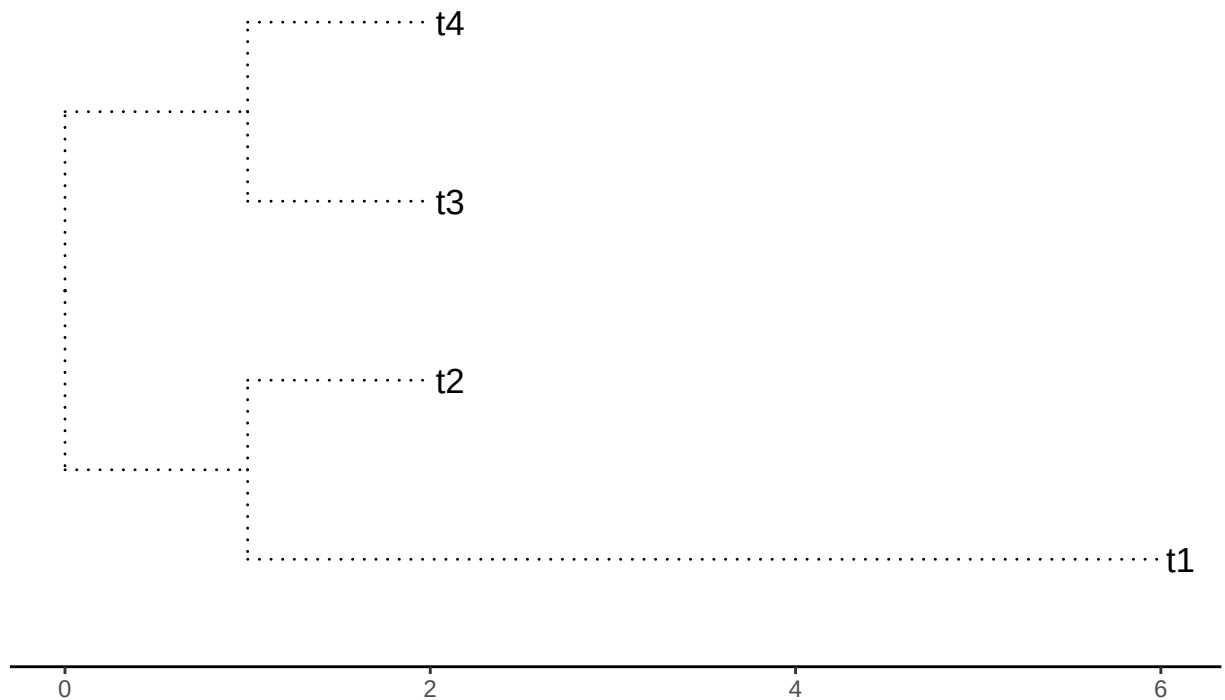
initialTree$edge.length[2] <- 5

plotTree <- ggtree(initialTree, ladderize=TRUE,
                   color="black", size=0.5, linetype="dotted") +
  geom_tiplab(size=4.5, color="black") +
  theme_tree2() +
  ggtitle(paste("Four terminals. Branches 1=",initialTree$edge.length[1],
               ". 2=",initialTree$edge.length[2],".",sep=""))

##
print(plotTree)

```

Four terminals. Branches 1=1. 2=5.



```
for( modelo in c("simpleswap","allswap","uniform") ){
  for( rama in c("terminals","internals","all") ){
    ## cat( "\n\t Branchs swapped=\t",rama,".\t",sep="" )
    val <- swapBL(tree=initialTree,
                  distribution = dist4taxa,
                  model = modelo,
                  branch = rama
                )
    printSwapBL(val, compact=TRUE)

    cat( "\n\n" )
  }
}
```

```
## model to test simpleswap reps 100
##   A1 A2
## 1 64 36
##
##
## model to test simpleswap reps 100
##   A1
## 1 100
```

```

##
##
## model to test simpleswap reps 100
##   A1 A1A2 A2
## 1 71   15 14
##
##
## model to test allswap reps 100
##   A1 A2
## 1 56 44
##
##
## model to test allswap reps 100
##   A1
## 1 100
##
##
## model to test allswap reps 100
##   A1 A1A2 A2
## 1 45   22 33
##
##
## model to test uniform reps 100
##   A1 A2
## 1 59 41
##
##
## model to test uniform reps 100
##   A1
## 1 100
##
##
## model to test uniform reps 100
##   A1 A2
## 1 44 56

AllTerminals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="terminals",
                           approach="all")

printEvalBranch(AllTerminals, compact=FALSE)

##   node initialArea FinalArea Aproach  %Delta
## 1    1          A1      A2   lower -80.193
## 2    2          A1      A1   lower      0
## 3    3          A1      A1   lower      0
## 4    4          A1      A1   lower      0
## 5    1          A1      A1   upper      0
## 6    2          A1      A2   upper 400.005
## 7    3          A1      A1   upper      0
## 8    4          A1      A2   upper 400.005

## Now the internals

```

```

AllInternals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="internals",
                           approach="all")

printEvalBranch(AllInternals, compact=FALSE)

##   node initialArea FinalArea Aproach %Delta
## 1     6           A1         A1  lower     0
## 2     7           A1         A1  lower     0
## 3     6           A1         A1  upper     0
## 4     7           A1         A1  upper     0
#now with a longer branch

initialTree <- tree

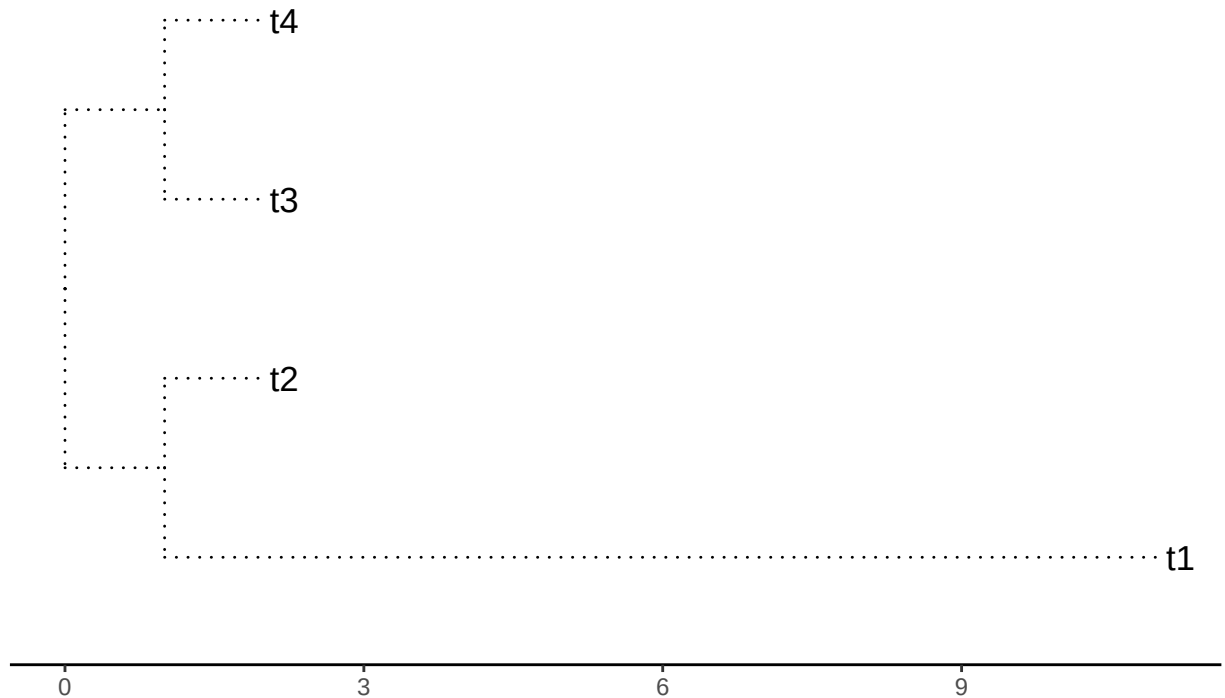
initialTree$edge.length[2] <- 10

plotTree <- ggtree(initialTree, ladderize=TRUE,
                   color="black", size=0.5, linetype="dotted") +
  geom_tiplab(size=4.5, color="black") +
  theme_tree2() +
  ggtitle(paste("Four terminals. Branches 1=",initialTree$edge.length[1],
                ". 2=",initialTree$edge.length[2],".",sep=""))

##
print(plotTree)

```

Four terminals. Branches 1=1. 2=10.



```
for( modelo in c("simpleswap","allswap","uniform") ){
  for( rama in c("terminals","internals","all") ){
    ## cat( "\n\t Branchs swapped=\t",rama,".\t",sep="" )
    val <- swapBL(tree=initialTree,
                  distribution = dist4taxa,
                  model = modelo,
                  branch = rama
                )
    printSwapBL(val, compact=TRUE)

    cat( "\n\n" )
  }
}
```

```
## model to test simpleswap reps 100
##   A1 A2
## 1 70 30
##
##
## model to test simpleswap reps 100
##   A1
## 1 100
```

```

##
##
## model to test simpleswap reps 100
##   A1 A1A2 A2
## 1 80   7 13
##
##
## model to test allswap reps 100
##   A1 A2
## 1 45 55
##
##
## model to test allswap reps 100
##   A1
## 1 100
##
##
## model to test allswap reps 100
##   A1 A1A2 A2
## 1 29   34 37
##
##
## model to test uniform reps 100
##   A1 A2
## 1 45 55
##
##
## model to test uniform reps 100
##   A1
## 1 100
##
##
## model to test uniform reps 100
##   A1 A2
## 1 52 48

AllTerminals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="terminals",
                           approach="all")

printEvalBranch(AllTerminals, compact=FALSE)

##   node initialArea FinalArea Aproach  %Delta
## 1    1          A1      A2   lower -90.096
## 2    2          A1      A1   lower    0
## 3    3          A1      A1   lower    0
## 4    4          A1      A1   lower    0
## 5    1          A1      A1   upper    0
## 6    2          A1      A2   upper 900.004
## 7    3          A1      A1   upper    0
## 8    4          A1      A2   upper 900.004

## Now the internals

```

```

AllInternals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="internals",
                           approach="all")

printEvalBranch(AllInternals, compact=FALSE)

##   node initialArea FinalArea Aproach %Delta
## 1     6           A1         A1  lower     0
## 2     7           A1         A1  lower     0
## 3     6           A1         A1  upper     0
## 4     7           A1         A1  upper     0
#now with a longer branch

initialTree <- tree

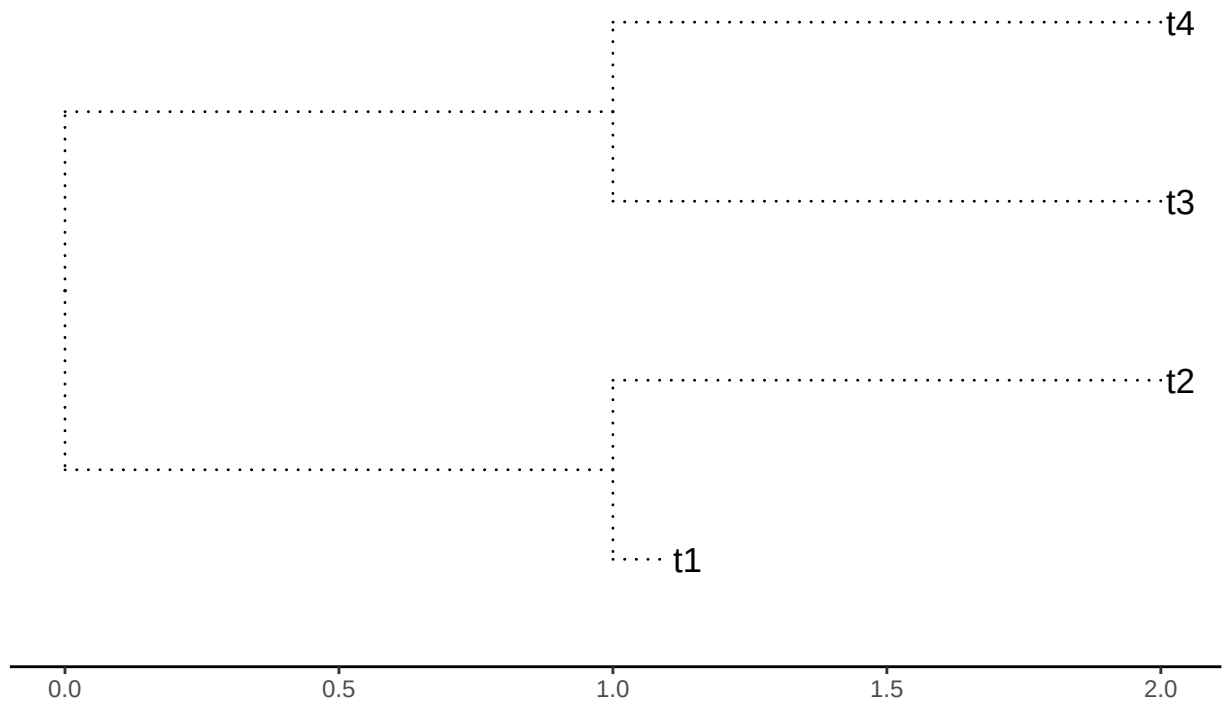
initialTree$edge.length[2] <- 0.1

plotTree <- ggtree(initialTree, ladderize=TRUE,
                   color="black", size=0.5, linetype="dotted") +
  geom_tiplab(size=4.5, color="black") +
  theme_tree2() +
  ggtitle(paste("Four terminals. Branches 1=",initialTree$edge.length[1],
               ". 2=",initialTree$edge.length[2],".",sep=""))

##
print(plotTree)

```

Four terminals. Branches 1=1. 2=0.1.



```
for( modelo in c("simpleswap","allswap","uniform") ){
  for( rama in c("terminals","internals","all") ){
    ## cat( "\n\t Branchs swapped=\t",rama,".\t",sep="" )
    val <- swapBL(tree=initialTree,
                  distribution = dist4taxa,
                  model = modelo,
                  branch = rama
                )
    printSwapBL(val, compact=TRUE)

    cat( "\n\n" )
  }
}
```

```
## model to test simpleswap reps 100
##   A1 A2
## 1 27 73
##
##
## model to test simpleswap reps 100
##   A2
## 1 100
```

```

##
##
## model to test simpleswap reps 100
##   A1 A1A2 A2
## 1 14   12 74
##
##
## model to test allswap reps 100
##   A1 A2
## 1 46 54
##
##
## model to test allswap reps 100
##   A2
## 1 100
##
##
## model to test allswap reps 100
##   A1 A1A2 A2
## 1 38   37 25
##
##
## model to test uniform reps 100
##   A1 A2
## 1 52 48
##
##
## model to test uniform reps 100
##   A2
## 1 100
##
##
## model to test uniform reps 100
##   A1 A2
## 1 49 51

AllTerminals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="terminals",
                           approach="all")

printEvalBranch(AllTerminals, compact=FALSE)

##   node initialArea FinalArea Aproach  %Delta
## 1    1          A2      A2   lower      0
## 2    2          A2      A1   lower -90.115
## 3    3          A2      A2   lower      0
## 4    4          A2      A1   lower -90.115
## 5    1          A2      A1   upper 900.408
## 6    2          A2      A2   upper      0
## 7    3          A2      A1   upper  90.005
## 8    4          A2      A2   upper      0

## Now the internals

```

```
AllInternals <- evalBranch(tree=initialTree,  
                           distribution = dist4taxa,  
                           branchToEval="internals",  
                           approach="all")
```

```
printEvalBranch(AllInternals, compact=FALSE)
```

##	node	initialArea	FinalArea	Approach	%Delta
## 1	6	A2	A2	lower	0
## 2	7	A2	A2	lower	0
## 3	6	A2	A2	upper	0
## 4	7	A2	A2	upper	0